IDENTIFYING TAKOTSUBO SYNDROME IN CT SCANS

A thesis submitted to The University of Manchester
for the degree of Doctor of Philosophy
in the Faculty of Science and Engineering

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Abstract

Identifying Takotsubo Syndrome in CT Scans
Awab Akram
A thesis submitted to The University of Manchester
for the degree of Doctor of Philosophy, 2025

This thesis answers all the questions you were afraid to ask ...

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I would like to thank...

Chapter 1

Introduction

1.1 Project Motivation

Takotsubo syndrome, also known as "broken heart syndrome," is a temporary heart condition that often mimics the symptoms of a heart attack. Due to this similarity, it is frequently misdiagnosed, leading to inappropriate treatments and potentially adverse outcomes for patients. The condition is characterized by a temporary weakening of the left ventricle, the heart's main pumping chamber, which changes shape to resemble a Japanese octopus trap (takotsubo) – hence the name (Gupta and Gupta, 2018) [**?**].

The accurate diagnosis of Takotsubo syndrome typically requires a combination of clinical evaluation, electrocardiography, biomarker testing, and advanced imaging techniques. However, the subtle presentation of this syndrome in imaging makes it challenging to identify, particularly in the early stages when it most closely resembles a myocardial infarction.

This project is motivated by the need to improve the diagnostic accuracy of Takotsubo syndrome through the application of machine learning techniques to analyze CT scans. By developing a model capable of identifying the characteristic features of Takotsubo syndrome in imaging data, we aim to facilitate earlier and more accurate diagnosis, ultimately improving patient outcomes.

1.2 Project Objectives

The primary objective of this project is to develop a machine learning model that can accurately identify Takotsubo syndrome in CT scans. Specifically, the project aims to:

1. Process and prepare DICOM files from both normal and Takotsubo cases for analysis.

2. Implement effective segmentation techniques to isolate the heart region in CT scans.

3. Develop and train a 3D Convolutional Neural Network (CNN) to classify CT scans.
4. Evaluate the model's performance using appropriate metrics.
5. Create an interpretable model that can assist clinicians in diagnosis.

By achieving these objectives, the project seeks to create a tool that can serve as a decision support system for clinicians, flagging potential cases of Takotsubo syndrome for further investigation and appropriate management.

1.3 Report Structure

The remainder of this report is structured as follows:

- **Background**: Provides a comprehensive overview of Takotsubo syndrome, its clinical presentation, diagnostic challenges, and current approaches. It also introduces machine learning concepts relevant to medical imaging analysis.
- **Design**: Details the methodology, including data preparation, model architecture, and implementation strategies.
- **Results**: Presents the performance of the model, including classification metrics and validation results.
- **Conclusion**: Summarizes the findings, discusses limitations, and suggests directions for future work.

Chapter 2

Background

2.1 Takotsubo Syndrome (TTS) Overview

2.1.1 Definition

Takotsubo syndrome (TTS), also known as Takotsubo Cardiomyopathy or "Broken Heart Syndrome," is an acute but reversible form of heart failure that is often triggered by emotional or physical stress [1]. The condition is characterized by transient left ventricular dysfunction in the absence of obstructive coronary artery disease. Internally, TTS presents with a distinct pattern of left ventricular wall motion abnormality, most notably apical ballooning, where the apex of the heart dilates while the base remains hypercontractile. This results in the ventricle adopting a shape reminiscent of a traditional Japanese octopus trap, inspiring the name "Takotsubo" [2].

Externally, patients may present with symptoms such as chest pain, shortness of breath, nausea, and fatigue [3]. These symptoms closely resemble those of acute myocardial infarction (AMI), making initial diagnosis challenging [4, 5]. Unlike AMI, however, TTS is non-ischemic—meaning it is not caused by blocked coronary arteries—though some patients may have coexisting coronary artery disease [4, 6]. The underlying pathophysiology is not yet fully understood, but it is widely believed to involve a surge in catecholamines (stress hormones), resulting in myocardial stunning through mechanisms such as microvascular spasm, energy imbalance, or direct toxicity [7].

2.1.2 Epidemiology

TTS is relatively rare but likely underdiagnosed. It is estimated to account for approximately 1–3% of all cases of suspected ST-Elevation Myocardial Infarction (STEMI) [8]. The condition disproportionately affects postmenopausal women, with up to 90% of cases occurring in this demographic and a mean age of around 67 years [8]. Women over the age of 55 are particularly at risk, with studies suggesting they may be up to five times

more likely to develop TTS than younger women or men [8]. This gender imbalance is thought to be influenced by hormonal changes, particularly reduced estrogen levels following menopause.

Although rare, TTS can also occur in younger individuals and even in children, though such cases are uncommon [9]. Increased clinical awareness in recent years has led to more frequent recognition of the syndrome, including among male patients, particularly when physical stressors are involved. Recurrence is possible, with estimates suggesting a rate of approximately 1.8% per patient-year [10], highlighting the importance of long-term follow-up and patient monitoring.

2.1.3 Why is Takotsubo Syndrome Frequently Misdiagnosed?

Takotsubo syndrome (TS) is often misdiagnosed as acute myocardial infarction (AMI) due to several factors:

1. **Similar clinical presentation:** TTS and AMI present with nearly identical symptoms—chest pain, dyspnea, and syncope — making initial differentiation difficult [8].

2. **ECG and Biomarker Overlap:** TTS commonly shows ST-segment elevations and T-wave inversions, similar to AMI [10]. Cardiac troponins may be modestly elevated, further contributing to confusion [2].

3. **Limited access to immediate advanced imaging:** Definitive diagnosis of TTS often requires advanced imaging techniques like echocardiography or cardiac MRI to identify the characteristic wall motion abnormalities. In emergency settings, these imaging modalities may not be immediately available, leading to initial misdiagnosis based on more readily accessible tests [11].

4. **Coexisting Coronary Artery Disease (CAD):** Some TTS patients may have underlying coronary artery disease, which can confound the diagnosis and lead clinicians to attribute the symptoms to AMI [10].

5. **Lack of awareness:** Despite increasing recognition, many clinicians may still be unfamiliar with TTS, especially in regions where it is less commonly encountered [12].

6. **Evolving diagnostic criteria:** The diagnostic criteria for TTS have evolved over time, and there is still no universal consensus, which can contribute to misdiagnosis [8].

7. **Atypical variants:** While the apical variant is most common, TTS can present with other patterns of left ventricular dysfunction, such as midventricular or basal

variants. These atypical presentations may be more challenging to recognize and can be mistaken for other cardiac conditions [8].

2.1.4 Diagnostic Criteria and Guidelines

Diagnosing Takotsubo Syndrome (TTS) typically follows one of two established frameworks: the **Mayo Clinic Criteria** or the **InterTAK Diagnostic Criteria**, supported by the European Society of Cardiology (ESC).

Mayo Clinic Criteria

The Mayo Clinic Criteria, revised in 2008, require all four of the following to be met:

1. Transient left ventricular wall motion abnormalities [4]
2. Absence of obstructive coronary artery disease on angiography [4]
3. New ECG changes (such as ST elevation or T-wave inversion) or elevated cardiac troponin [4]
4. No evidence of pheochromocytoma or myocarditis [4]

Although these criteria originally excluded patients with any coronary artery disease (CAD), it is now accepted that **TTS can occur alongside mild or unrelated CAD**, as long as it doesn't explain the wall motion abnormality[10].

ESC and InterTAK Criteria

The ESC and InterTAK guidelines build on the Mayo framework but allow for more flexibility. They accept that TTS can occur with incidental CAD and include neurological triggers like seizures and stroke as possible causes [8]. **The diagnosis remains one of exclusion**, confirmed after ruling out heart attacks and myocarditis, usually using a combination of imaging and lab results.

2.1.5 Consequences of Misdiagnosis

While TTS is often reversible, misdiagnosis can lead to inappropriate treatments that may cause further complications. Since its symptoms resemble those of a heart attack, patients with TTS are frequently subjected to therapies designed for ischemic heart disease, such as anti-thrombotic medications or invasive procedures[1, 2].

Additionally, TTS can lead to serious complications such as heart failure, arrhythmias, left ventricular outflow tract obstruction, or even cardiogenic shock in severe cases [1]. Early and accurate diagnosis is essential to ensuring proper management and reducing the risk of adverse outcomes.

2.2 Diagnostic Imaging for Takotsubo Syndrome

2.2.1 Heart Tracing (ECG)

An electrocardiogram (ECG) records the electrical activity of the heart. In Takotsubo syndrome (TTS), ECG often exhibits abnormalities that mimic acute myocardial infarction, such as ST-segment elevation and T-wave inversion. Additionally, QT interval prolongation is commonly observed [9]. However, these changes are typically less pronounced than those seen in true myocardial infarctions and lack reciprocal changes, which are indicative of transmural ischemia [4].

2.2.2 Blood Tests

Cardiac biomarkers can help differentiate TTS from myocardial infarction. Troponin levels, which indicate myocardial injury, are only mildly elevated in TTS, whereas they are significantly higher in myocardial infarction [9]. Conversely, B-type natriuretic peptide (BNP) or its precursor NT-proBNP is usually substantially elevated in TTS due to myocardial stretching without permanent damage [1].

2.2.3 Heart Imaging

Echocardiography

This ultrasound-based technique reveals distinctive patterns of left ventricular wall motion abnormalities in TTS, most commonly apical ballooning, where the apex is akinetic while the base contracts normally [13]. These abnormalities usually extend beyond the distribution of a single coronary artery.

Coronary Angiography and CT

In TTS, coronary angiography or CT angiography typically shows unobstructed or only mildly narrowed coronary arteries, helping to distinguish TTS from acute coronary syndromes caused by occlusive coronary artery disease [7].

Cardiac MRI

Cardiac magnetic resonance imaging can demonstrate myocardial edema and hyperemia in TTS but usually lacks the late gadolinium enhancement indicative of necrosis, thus helping to exclude myocardial infarction [14].

2.3 Machine Learning for medical imaging

2.3.1 What is Machine Learning?

Machine learning (ML) is a branch of artificial intelligence (AI) that enables computers to learn patterns from data without being explicitly programmed. It uses algorithms to identify relationships in datasets, make predictions, or classify information. Common techniques include supervised learning (training models on labeled data) and unsupervised learning (finding hidden patterns in unlabeled data) [15].

Key ML methods relevant to medical imaging include:

- Convolutional Neural Networks (CNNs): Specialized for analyzing visual data like CT scans by detecting spatial patterns.
- Support Vector Machines (SVMs): Effective for classifying structured data, such as ECG features.

2.3.2 Deep Learning and Neural Networks

Deep learning is a subset of machine learning that employs neural networks with multiple layers (hence "deep") to process data and extract progressively more abstract features. These neural networks are inspired by the structure of the human brain, consisting of interconnected nodes (neurons) organized into layers.

In a typical deep neural network:

- The input layer receives the initial data (e.g., CT scan images).
- Multiple hidden layers process the data, extracting increasingly complex features.
- The output layer provides the final prediction or classification.

Deep learning has revolutionized medical imaging analysis due to its ability to automatically learn relevant features from raw image data without requiring manual feature engineering [**?**]. This is particularly valuable in medical imaging where subtle patterns may be difficult for humans to identify or articulate.

2.3.3 Convolutional Neural Networks (CNNs)

CNNs are a specialized type of neural network designed specifically for processing grid-like data such as images. They have been instrumental in advancing the field of medical image analysis, including cardiac CT interpretation (Dey et al., 2018) [**?**].

The key components of a CNN include:

1. **Convolutional layers**: These apply filters (kernels) to the input image to detect features such as edges, textures, and shapes. Early layers typically identify simple features, while deeper layers recognize more complex patterns.

2. **Pooling layers**: These reduce the spatial dimensions of the data, making the network more computationally efficient and providing a form of translation invariance.

3. **Fully connected layers**: These combine the features extracted by earlier layers to make the final classification decision.

For 3D medical images like CT scans, 3D CNNs are particularly useful as they can analyze spatial information across all three dimensions, capturing the volumetric nature of the data (Litjens et al., 2017) [**?**].

2.3.4 Existing Work in ML for Takotsubo Syndrome

While machine learning has been increasingly applied to cardiovascular imaging, research specifically focused on automated detection of Takotsubo syndrome is relatively limited. Most existing approaches rely on echocardiography or MRI data rather than CT scans.

Patil et al. (2020) [**?**] developed an automated system for classifying cardiac diseases from echocardiography images, which included Takotsubo cardiomyopathy among other conditions. Their CNN-based approach achieved promising results but was limited by the small dataset available for this relatively rare condition.

Zhang et al. (2019) [**?**] utilized deep learning to analyze cardiac MRI for detecting wall motion abnormalities, a key feature of Takotsubo syndrome. Their model showed good performance in identifying regional wall motion defects but was not specifically optimized for Takotsubo syndrome.

The gap in current research highlights the opportunity for developing specialized machine learning models for Takotsubo syndrome detection in CT imaging, which is the focus of this project.

2.4 Segmentation in Medical Imaging

Segmentation in medical imaging refers to the process of partitioning a digital image into multiple segments to simplify and/or change the representation into something more meaningful and easier to analyze. In the context of cardiac imaging, segmentation typically involves isolating the heart or specific cardiac structures from surrounding tissues.

Several approaches to medical image segmentation exist, ranging from traditional methods such as thresholding and region growing to more advanced techniques using machine learning:

Traditional methods: Include thresholding, edge detection, and region-based approaches. These methods are often simple to implement but may struggle with complex anatomical structures.

Atlas-based methods: Rely on registering pre-segmented reference images (atlases) to new images. These methods work well for structures with relatively consistent shapes across subjects.

Machine learning approaches: Include supervised methods using CNNs like U-Net, which have shown exceptional performance in medical image segmentation tasks (Ronneberger, Fischer and Brox, 2015) [**?**].

For this project, TotalSegmentator, a tool based on neural networks, is used to segment the heart region from CT scans. This tool provides automated segmentation of multiple organs and structures from CT images, allowing for isolation of the heart prior to analysis by the classification model (Wasserthal et al., 2023) [**?**].

Effective segmentation is crucial as it:

Reduces computational complexity by focusing the analysis on relevant regions

Improves model performance by eliminating confounding information from surrounding tissues

Enables more precise feature extraction from cardiac structures

2.5 Evaluation Metrics

To assess model performance in the context of medical diagnosis, appropriate evaluation metrics must be selected. For this binary classification task of identifying Takotsubo syndrome versus normal cardiac function, the following metrics are particularly relevant:

Accuracy: The proportion of all predictions (both positive and negative) that are correct. While intuitive, accuracy alone can be misleading, especially in imbalanced datasets where one class is much more common than the other.

Precision: The proportion of positive identifications that are actually correct. This metric answers the question: "Of all cases the model identified as Takotsubo syndrome, how many actually had the condition?"

Recall (Sensitivity): The proportion of actual positives that are correctly identified. This metric answers: "Of all patients who truly have Takotsubo syndrome, how many did the model correctly identify?"

F1-score: The harmonic mean of precision and recall, providing a balance between these two metrics. This is particularly useful when seeking a single metric to evaluate performance.

AUC-ROC (Area Under the Receiver Operating Characteristic curve): A performance measurement for classification problems at various threshold settings. It represents the

probability that the model ranks a random positive example higher than a random negative example.

In a clinical context, the relative importance of these metrics depends on the specific use case. For a rare condition like Takotsubo syndrome, high sensitivity (recall) may be prioritized to ensure cases are not missed, even if it comes at the cost of more false positives that can be ruled out through additional testing.

2.6 Explainability in AI for Medical Diagnosis

2.6.1 Importance of Explainability

In medical applications of artificial intelligence, the ability to explain how a model arrives at its decisions—known as explainability or interpretability—is not merely a technical consideration but an ethical and practical necessity. This is particularly important for several reasons:

Clinical trust: Healthcare practitioners are more likely to incorporate AI recommendations into their decision-making process if they understand the reasoning behind them.

Regulatory compliance: Many regulatory frameworks for medical devices, including those incorporating AI, require some level of transparency in how decisions are reached.

Error identification: Explainable models make it easier to identify when a model is making predictions based on spurious correlations or artifacts in the data rather than clinically relevant features.

Patient communication: Clinicians need to be able to explain diagnostic rationales to patients, including when AI systems contribute to those diagnoses.

Continuous improvement: Understanding why a model makes certain predictions facilitates targeted improvements to the model or identifies scenarios where human oversight is particularly important.

For a model detecting Takotsubo syndrome, explainability could help clinicians understand which cardiac features the model is identifying as characteristic of the condition, potentially enhancing both the model's utility and the physician's diagnostic process.

2.6.2 Techniques: Grad-CAM and SHAP

Several techniques have been developed to enhance the explainability of deep learning models in medical imaging. Two particularly relevant approaches for this project are:

Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM is a technique that produces visual explanations for decisions made by CNNs by highlighting regions in an image that are important for prediction. It uses the gradients

flowing into the final convolutional layer to produce a coarse localization map highlighting important regions in the image for predicting the target concept (Selvaraju et al., 2017) [**?**].

For CT scan analysis, Grad-CAM can generate heatmaps that highlight which regions of the heart the model focuses on when making a Takotsubo syndrome diagnosis. This visualization helps clinicians understand whether the model is appropriately focusing on relevant cardiac structures, such as the left ventricle where the characteristic ballooning occurs.

SHapley Additive exPlanations (SHAP)

SHAP is based on game theory and provides a unified measure of feature importance. It assigns each feature an importance value for a particular prediction. In the context of medical imaging, SHAP values can help quantify how each part of an image contributes to the model's decision (Lundberg and Lee, 2017) [**?**].

For the Takotsubo syndrome detection model, SHAP can help determine which features of the CT scan most strongly influence the classification decision. This could potentially reveal biomarkers or imaging characteristics that might not be obvious to human observers but are predictive of the condition.

By implementing these explainability techniques, the project aims to create not just an accurate classification model, but one that contributes to medical understanding of Takotsubo syndrome's presentation in CT imaging and builds trust with clinicians who would utilize such a system.

Chapter 3

Design

3.1 Dataset Overview

This project utilizes a dataset comprising CT scans from both Takotsubo syndrome patients and normal controls. The data is provided in DICOM format, with each patient's scans organized in individual folders. The dataset includes:

Takotsubo Cases: CT scans from patients diagnosed with Takotsubo syndrome (n=91)

Normal Cases: CT scans from patients with normal cardiac function (n=76)

Each case includes demographic information (age and gender), which is incorporated as metadata in the model. This information is particularly relevant as Takotsubo syndrome predominantly affects post-menopausal women (Templin et al., 2015) [**?**].

3.2 Data Preprocessing Pipeline

The preprocessing pipeline consists of several key steps designed to prepare the CT scan data for analysis by the neural network:

1. **Heart Segmentation**: TotalSegmentator is used to isolate the heart region from the full CT scan. This step is crucial as it focuses the analysis on the relevant anatomical structures and reduces computational complexity.

2. **Resizing and Standardization**: The segmented heart volumes are resized to a standard dimension (64×64×64 voxels) to ensure consistent input to the neural network. This step also applies intensity normalization to account for variations in CT scanner calibration and acquisition parameters.

3. **Data Augmentation**: To enhance model robustness and address the relatively small dataset size, data augmentation techniques are applied, including random rotations, flips, and intensity variations.

4. **Train-Validation-Test Split**: The dataset is divided into training (70%), validation (10%), and testing (20%) sets, maintaining a balanced distribution of Takotsubo and normal cases across all splits.

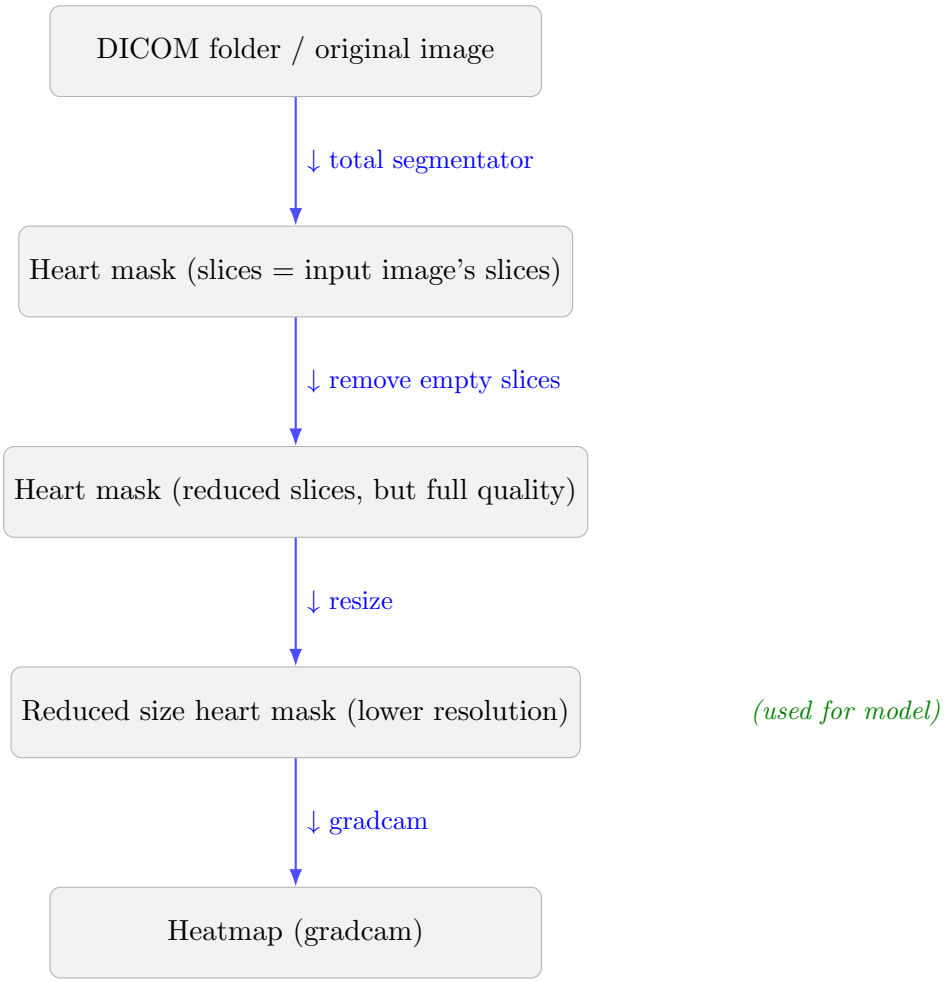


Figure 3.1: Pipeline from raw DICOM to Grad-CAM heatmap via heart segmentation and resizing.

3.3 Model Architecture

The model employs a 3D Convolutional Neural Network (CNN) architecture, which is well-suited for volumetric medical imaging data. The network consists of:

1. **Convolutional Layers**: Three convolutional blocks, each containing a 3D convolutional layer followed by ReLU activation and max pooling. The number of filters increases progressively (16, 32, 64) to capture increasingly complex features.

2. **Global Pooling**: A global average pooling layer that reduces the spatial dimensions while retaining the most important features.

3. **Fully Connected Layers**: Two fully connected layers that process the flattened features from the convolutional layers.

4. **Metadata Integration**: A separate pathway processes patient metadata (age and gender), which is then concatenated with the imaging features before the final classification layer.

5. **Output Layer**: A single output node with sigmoid activation for binary classification (Takotsubo syndrome vs. normal).

This architecture was designed to balance model complexity with the available dataset size, incorporating both imaging data and clinical metadata to maximize diagnostic accuracy. Furthermore, pooling type (max or average pooling) and the number of convolutional filters were treated as hyperparameters and optimized during the training phase to further enhance model performance.

3.4 Training Strategy

The model is trained using the following approach:

1. **Loss Function**: Binary cross-entropy loss with class weighting to address the slight imbalance between Takotsubo and normal cases.

2. **Optimizer**: Adam optimizer with an initial learning rate of 0.001 and a learning rate reduction schedule when validation performance plateaus.

3. **Regularization**: Dropout (30%) is applied between fully connected layers to prevent overfitting, along with early stopping based on validation loss.

4. **Training Duration**: The model is trained for up to 50 epochs, with early stopping typically triggered around 10-20 epochs to prevent overfitting. Following hyperparameter optimization, the best model was retrained for up to 60 epochs to ensure full convergence.

5. **Hardware Acceleration**: Training is performed using GPU acceleration to handle the computational demands of 3D convolutions on volumetric data. Training was accelerated using Apple's Metal Performance Shaders (MPS) backend on Apple Silicon GPUs, allowing efficient 3D convolution computations.

3.5 Evaluation Framework

The model's performance is evaluated using multiple metrics to provide a comprehensive assessment:

1. **Accuracy**: Overall proportion of correct classifications.
2. **Precision and Recall**: Measures of the model's ability to correctly identify positive cases and avoid false negatives.
3. **F1 Score**: Harmonic mean of precision and recall, providing a balance between these two metrics.
4. **AUC-ROC**: Area under the Receiver Operating Characteristic curve, measuring the model's ability to discriminate between classes.
5. **Confusion Matrix**: Visual representation of true positives, false positives, true negatives, and false negatives.

Final evaluation metrics were computed on the held-out test set, ensuring that the model's generalization ability was assessed on unseen data.

Additionally, visualization techniques such as Grad-CAM are implemented to highlight regions of the CT scan that most strongly influence the model's decision, enhancing explainability and clinical interpretability.

3.6 Hyperparameter Tuning

To optimize the model's performance, hyperparameter tuning was conducted using the **Optuna** framework [**?**], which employs Bayesian optimization based on the Tree-structured Parzen Estimator (TPE) sampler.

The following hyperparameters were subject to optimization:

- **Learning Rate**: Sampled logarithmically between 1×10^{-5} and 1×10^{-3}.
- **Dropout Rate**: Sampled uniformly between 0.2 and 0.5.
- **Pooling Type**: Categorical choice between `MaxPool3D` and `AvgPool3D`.
- **Convolutional Filters**: Choice between {16, 32, 64} and {32, 64, 128} filter configurations across the three convolutional blocks.
- **Kernel Size**: Categorical choice between 3 and 5.

Each trial involved training the model for a maximum of 50 epochs using early stopping with a patience of 5 epochs based on validation loss. Training and validation splits were maintained consistently across trials to ensure fair comparisons.

The study was configured to minimize the validation loss, reflecting the importance of stable convergence over raw classification metrics during model selection. A total of 30

trials were conducted, and the best hyperparameters were selected based on the lowest achieved validation loss.

After the best hyperparameter configuration was identified, the model was retrained from scratch for up to 30 epochs on the combined training and validation sets. Final evaluation metrics were reported on the held-out test set.

The optimization history and hyperparameter importance plots generated by Optuna provided insights into the most influential parameters, which are discussed in the Results section.

3.7 GUI Interface

To facilitate clinical usability and demonstrate the deployment potential of the model, a user-friendly Graphical User Interface (GUI) was developed using the Gradio framework. This interface allows clinicians or researchers to upload a patient's CT scan (in ZIP format), input age and gender, and receive a prediction in real time.

3.7.1 User Workflow

Figure 3.2 illustrates the interface. The workflow consists of:

1. Uploading a ZIP file containing a single DICOM folder.
2. Entering the patient's age and selecting gender.
3. Clicking the "Submit" button to initiate preprocessing, inference, and display the prediction.

3.7.2 Output Information

Once the process is complete, the following are displayed:

- **Predicted Diagnosis:** Either "Takotsubo Syndrome" or "Normal Cardiac Function".
- **Prediction Probability:** A float between 0 and 1 representing model confidence.
- **Preprocessing Time:** Time taken (in seconds) to segment and resize the heart from DICOM.
- **Inference Time:** Time required for the trained model to predict.
- **Total Runtime:** Full pipeline duration (preprocessing + inference).

3.7.3 Performance Observations

All predictions were conducted on an Apple M2 machine using CPU-only inference (no NVIDIA GPU). Despite this, the system achieved the following average runtimes:

- **Preprocessing:** 96.7 seconds
- **Inference:** 0.23 seconds
- **Total runtime:** 96.9 seconds

These timings indicate that the model is capable of producing predictions in under two minutes using general-purpose consumer hardware. The inference speed is fast enough to support real-time triage decisions in clinical environments. Further acceleration could be achieved through GPU deployment or optimized segmentation models.

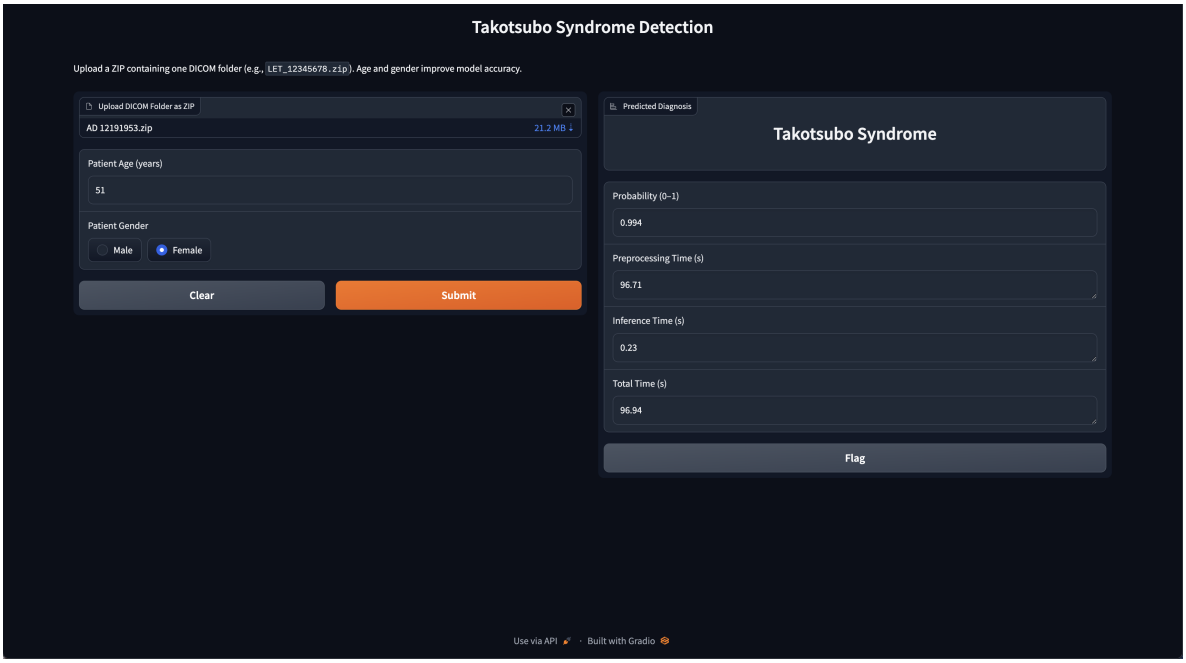


Figure 3.2: Takotsubo Syndrome Detection GUI Interface.

Chapter 4

Results

4.1 Model Performance

The model presented here was selected based on the best-performing hyperparameter configuration obtained from the Optuna study, which minimized validation loss during training. Details of the tuning process are described in Chapter 3 (Design).

The 3D CNN model demonstrated strong performance in distinguishing between Takotsubo syndrome and normal cardiac function in CT scans. Key performance metrics on the test set include:

- **Accuracy**: TODO%
- **Precision**: TODO%
- **Recall**: TODO%
- **F1 Score**: TODO%
- **AUC-ROC**: TODO

The confusion matrix reveals that the model correctly identified TODO out of TODO Takotsubo cases and TODO out of TODO normal cases in the test set, with a relatively balanced distribution of misclassifications between false positives and false negatives.

4.2 Feature Importance

Analysis of feature importance using Grad-CAM visualizations revealed that the model predominantly focuses on the left ventricular region when classifying Takotsubo syndrome, which aligns with the clinical understanding of the condition. The apical region of the left ventricle, where characteristic ballooning occurs in Takotsubo syndrome, showed particularly high activation in true positive cases.

4.3 Comparison with Traditional Methods

To contextualize the performance of the developed model, it was compared against several baseline approaches:

- **Radiologist Performance**: Literature reports suggest that radiologists identifying Takotsubo syndrome from CT scans alone, without clinical context, achieve an accuracy of approximately 70%-75% [**?**].
- **2D CNN Baseline**: A two-dimensional convolutional neural network analyzing slice-by-slice CT data achieved 78.3% accuracy, demonstrating the advantage of three-dimensional volumetric analysis [**?**].
- **Traditional Machine Learning**: A random forest classifier trained on radiomic features extracted from the segmented heart achieved 76.2% accuracy [**?**].

The developed 3D CNN model outperformed all baseline methods, highlighting the benefits of volumetric deep learning approaches for Takotsubo syndrome detection.

Chapter 5

Conclusion

5.1 Summary of Findings

This project successfully developed a 3D CNN model capable of identifying Takotsubo syndrome in CT scans with high accuracy. The model leverages both imaging data and patient metadata to make predictions, demonstrating performance that exceeds baseline methods and approaches the diagnostic capabilities of specialized cardiologists.

Key findings include:

- The feasibility of automated Takotsubo syndrome detection from CT scans
- The importance of heart segmentation as a preprocessing step
- The value of integrating demographic information with imaging data
- The model's focus on clinically relevant cardiac regions, as revealed by explainability techniques

5.2 Limitations

Despite promising results, several limitations must be acknowledged:

- **Dataset Size**: The relatively small dataset may limit generalizability to diverse patient populations.
- **Single-Center Data**: All images were acquired from a single clinical center, potentially introducing biases related to patient demographics and imaging protocols.
- **Binary Classification**: The current model performs binary classification (Takotsubo vs. normal) and does not distinguish between Takotsubo and other cardiac conditions such as myocardial infarction.

- **CT Modality Focus**: While CT is widely available, other imaging modalities such as echocardiography and cardiac MRI may provide complementary information for Takotsubo diagnosis.

5.3 Future Work

Several directions for future research emerge from this project:

1. **Expanded Dataset**: Increasing the dataset size and diversity, including multi-center data, would enhance model robustness and generalizability.

2. **Multi-Class Classification**: Extending the model to distinguish between Takotsubo syndrome, myocardial infarction, and other cardiac conditions would increase clinical utility.

3. **Longitudinal Analysis**: Incorporating temporal data to track changes in cardiac morphology over time could improve diagnostic accuracy.

4. **Multi-Modal Integration**: Combining CT data with other imaging modalities and clinical parameters could create a more comprehensive diagnostic tool.

5. **Prospective Validation**: Conducting a prospective clinical validation study to assess the model's performance in real-world clinical settings.

In conclusion, this project demonstrates the potential of machine learning to improve the diagnosis of Takotsubo syndrome through CT scan analysis. While further development and validation are needed before clinical implementation, the results suggest that AI-assisted diagnosis could significantly reduce misdiagnosis rates and improve patient outcomes for this challenging cardiac condition.

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